

Aviation Risk Analysis: From Data to Defensible Decisions

A Quantitative Framework for Selecting Safer Operators

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Moringa School Phase 1 Project

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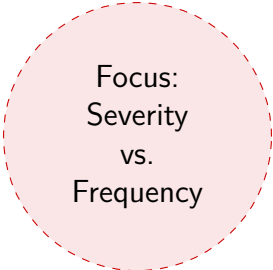
Overview: The Business Problem

The Cost of Uncertainty

Aviation accidents create massive business costs, including loss of life, liability, and reputational damage.

The Challenge:

- Vendor selection often relies on intuition or simple “accident counts.”
- This penalizes high-volume carriers (who report minor incidents daily) and misses true risks.



Focus:
Severity
vs.
Frequency

Data Understanding: Sourcing the Truth

Data Sources

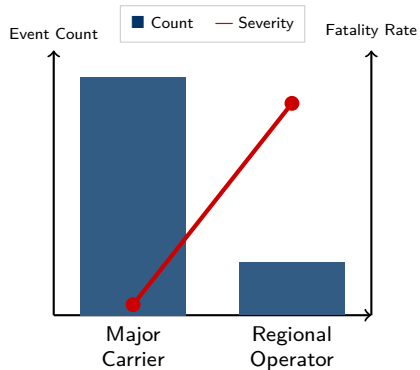
- **NTSB:** Detailed accident reports.
- **FAA:** Incident data (Frequency focus).
- **World Summary:** Global severity data.

Preparation Strategy

- **Standardization:** Cleaned operator names to group data correctly.
- **Filtering:** Excluded operators with < 5 events to ensure statistical stability.
- **Focus:** Distinguished between *Incidents* (minor) and *Accidents* (fatal).

Insight 1: Frequency $\neq$ Risk

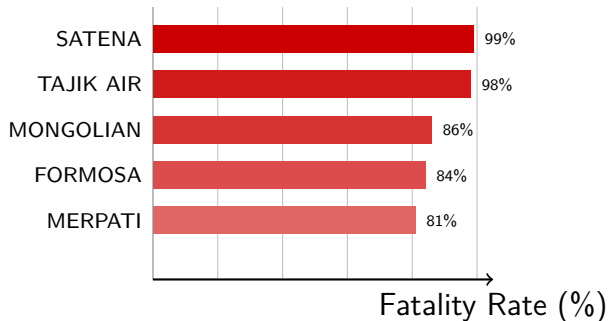
- **The Volume Trap:** Large carriers appear “unsafe” in raw counts simply because they fly thousands of times a day.
- **Reality:** The vast majority of these reports are non-fatal mechanical issues (Blue Bars).
- **True Risk:** Regional operators have fewer events, but higher severity (Red Line).



Insight 2: The “Weighted Fatality Rate”

Notebook Findings

- **Metric:** $\frac{\text{Total Fatalities}}{\text{Total Aboard}}$
- **Result:** Risk is concentrated in operators where accidents are rare but catastrophic.
- *Data Source: World Aircraft Accident Summary (df grouping analysis).*



Recommendations: A New Protocol



Action Item

Stop using raw accident counts. Adopt the **Weighted Fatality Rate** as the primary safety KPI.

The Benefit

Moves the business from a reactive posture (headlines) to a proactive safety gate (data).

Next Steps: Refining the Model

To improve precision and operationalize this model, I propose:

- **Integrate Exposure Data:** Acquire “Total Flight Hours” to calculate probability per flight (Risk vs. Severity).
- **Time-Windowing:** Weight recent data (last 5–10 years) more heavily to reflect modern fleet safety.
- **Dashboarding:** Build a live internal tool for the procurement team to check vendor safety ratings in real-time.

Thank You

“Moving from intuition to data-driven due diligence reduces liability and saves lives.”

Contact

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Resources

GitHub Repository
Jupyter Notebook